Godha Karthikeyan Yadav

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OBJECTIVE

B.Tech ECE graduate (2026) with strong software fundamentals in Python, C/C++, and SQL, backed by hands-on experience building computer-vision pipelines and full-stack data applications. Proven ability to design, test, and debug systems end-to-end. Seeking a Software Engineer role to build and ship reliable, well-tested software.

EDUCATION

Vardhaman College of Engineering — **Hyderabad, India**
B.Tech in Electronics and Communication Engineering (JNTUH) | CGPA: 7.3 — **2022 – 2026**

Sri Chaitanya Junior College — **Hyderabad, India**
Class 12th | Percentage: 88.2 — **2020 – 2022**

EXPERIENCE

Entuple Technologies – Design Verification Intern | Certificate | Project — **Oct 2024 – Nov 2024**
Bangalore, India

- Verified UART transmitter and receiver functionality using SystemVerilog testbenches, covering baud rate accuracy, data framing, and error handling.
- Simulated and debugged UART behavior using Cadence Xcelium, performing waveform-level analysis to correlate functional results with specification.
- Followed structured documentation and version control practices (Git) for testbench development and bug tracking.

PROJECTS

Retail Self-Checkout System using YOLOv10 | Project | IEEE Paper — **Mar 2025 – May 2025**

- Trained a custom YOLOv10 model in Python for real-time retail product detection and automated billing.
- Reduced checkout latency through model optimization and integration with a local product-price SQL database, automating regression testing across sample datasets.
- Published comparative benchmarking results (YOLOv10 vs. YOLOv11 vs. YOLOv12) at an IEEE conference.

Tech: *YOLOv10, Python, OpenCV, PyTorch, SQL*

LogSentry: AIOps Log Monitoring API | Project — **Jan 2026 – Feb 2026**

- Built a FastAPI backend that parses raw application logs, stores structured entries in MongoDB Atlas, computes rolling error-rate metrics, and triggers deduplicated threshold-based alerts with persistent history.
- Containerized the service with Docker, set up a GitHub Actions CI pipeline, and built a live auto-refreshing dashboard for real-time system health visibility.

Tech: *Python, FastAPI, MongoDB, Docker, REST APIs, GitHub Actions*

India Waste & Plastic Pollution Insights Dashboard | Project — **Dec 2024 – Sep 2025**

- Analyzed public solid-waste and plastic-pollution data across Indian states to identify processing gaps and high-risk regions.
- Built an interactive dashboard (state/year filters, choropleth map, trend charts) with a SQL backend and Python data pipeline, surfacing actionable recommendations for waste-management prioritization.

Tech: *SQL, Python, React, Data Visualization*

Image Classification on Remote Sensing Data using ResNet50 | Project — **Oct 2025 – Jan 2026**

- Built an end-to-end deep learning pipeline in Python for classifying SAR and multispectral satellite imagery using transfer learning on ResNet50V2, with a modular preprocessing/augmentation pipeline for high-dimensional remote sensing formats.

Tech: *Python, PyTorch, TensorFlow/Keras, OpenCV, Pandas, Scikit-learn*

TECHNICAL SKILLS

Languages: Python, C, C++, Java, SQL, Verilog, SystemVerilog
Software Development: REST API design, OOP, Data Structures & Algorithms, Git version control, Docker, GitHub Actions (CI), Postman API, unit testing
ML / Data: PyTorch, TensorFlow/Keras, OpenCV, Pandas, Scikit-learn, YOLOv10
Databases: MongoDB, SQL (joins, CTEs, window functions), MySQL, PostgreSQL
Web / Frontend: React, Data Visualization, FastAPI
Systems & Tools: Cadence Xcelium, Cadence Virtuoso, Xilinx Vivado, VS Code, QGIS
Coursework: Computer Architecture, OS, DSA, DBMS, OOPS

CERTIFICATIONS

- Oracle Cloud Infrastructure Certified AI Foundations Associate | Oracle University | Certificate
- Certified Associate Professional VLSI Design (ZC-920) | Zebrold | Certificate
- CUDA Programming | Pantech E-Learning (NVIDIA Jetson Nano) | Certificate
- 100 Days of Python | Udemy | Certificate
- Programming with Java | NPTEL | Certificate

ACHIEVEMENTS & ACTIVITIES

- Smart India Hackathon 2024 Qualified | Certificate • IEEE Publication: YOLOv10/11/12 Comparative Analysis for Retail Self-Checkout | Paper | Certificate